Module 3 STP and RSTP

Lab 3-1 Configuring STP

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Enable and disable STP.
- Change the STP mode that is used by a switch.
- Change the bridge priority to control root bridge election.
- Change the port priority to control election of the root port and designated port.
- Change the port cost to control election of the root port and designated port.
- Configure an edge port.

Topology

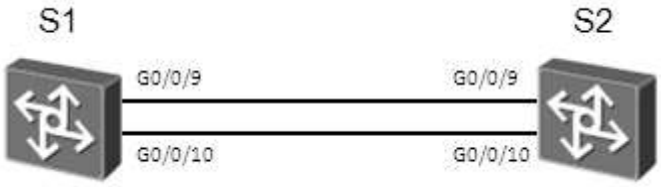


Figure 3.1 STP topology

Scenario

Assume that you are a network administrator of a company. The

company network consists of two layers: core layer and access layer. The network uses a design that supports network redundancy. STP will be used to prevent loops. The STP network should include setting the bridge priority to control STP root bridge election, and configuration of features to speed up STP route convergence.

Tasks

Step 1 Configure STP and verify the STP configuration.

Irrelevant interfaces must be disabled to ensure test result accuracy.

Shut down port interfaces Ethernet 0/0/1 on S3, Ethernet 0/0/13 and Ethernet 0/0/23 on S3, as well as Ethernet 0/0/14 and Ethernet 0/0/24 on S4 before starting STP configuration. Ensure that the devices start without any configuration files. If STP is disabled, run the **stp enable** command to enable STP.

```
<Quidway>system-view
[Quidway]sysname R3
[R3]interface Ethernet 0/0/1
[R3-Ethernet0/0/1]shutdown
[R3-Ethernet0/0/1]quit
[R3]interface Ethernet 0/0/13
[R3-Ethernet0/0/13]shutdown
[R3-Ethernet0/0/13]quit
[R3]interface Ethernet 0/0/23
[R3-Ethernet0/0/23]shutdown

<Quidway>system-view
[Quidway]sysname S4
[S4]inter Ethernet 0/0/14
[S4-Ethernet0/0/14]shutdown
[S4-Ethernet0/0/14]quit
[S4]interface Ethernet 0/0/24
[S4-Ethernet0/0/24]shutdown
```

In the lab,S1 and S2 are connected through two links, and STP is used. Enable STP on S1 and S2 and set S1 as the root.

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S1
```

```
[S1]stp mode stp
[S1]stp root primary

<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S2
[S2]stp mode stp
[S2]stp root secondary
```

Run the **display stp brief** command to view brief information about STP.

```
<S1>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    DESI    FORWARDING  NONE
    0       GigabitEthernet0/0/10   DESI    FORWARDING  NONE

<S2>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    ROOT    FORWARDING  NONE
    0       GigabitEthernet0/0/10   ALTE    DISCARDING  NONE
```

Run the **display stp interface** command to view the STP status of a port.

```
<S1>display stp interface GigabitEthernet 0/0/10
----[CIST][Port10(GigabitEthernet0/0/10)][FORWARDING]----
 Port Protocol      :Enabled
 Port Role          :Designated Port
 Port Priority      :128
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port   :0.4c1f-cc45-aace / 128.10
 Port Edged         :Config=default / Active=disabled
 Point-to-point     :Config=auto / Active=true
 Transit Limit      :147 packets/hello-time
 Protection Type    :None
 Port STP Mode      :STP
 Port Protocol Type  :Config=auto / Active=dot1s
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 20
 TC or TCN send     :17
 TC or TCN received  :33
 BPDU Sent          :221
        TCN: 0, Config: 221, RST: 0, MST: 0
```

```
 BPDU Received      :68
        TCN: 0, Config: 68, RST: 0, MST: 0
<S2>display stp interface GigabitEthernet 0/0/10
----[CIST][Port10(GigabitEthernet0/0/10)][DISCARDING]----
 Port Protocol      :Enabled
 Port Role          :Alternate Port
 Port Priority      :128
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port   :0.4c1f-cc45-aace / 128.10
 Port Edged         :Config=default / Active=disabled
 Point-to-point     :Config=auto / Active=true
 Transit Limit      :147 packets/hello-time
 Protection Type    :None
 Port STP Mode      :STP
 Port Protocol Type  :Config=auto / Active=dot1s
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 0
 TC or TCN send     :17
 TC or TCN received  :17
 BPDU Sent          :35
        TCN: 0, Config: 35, RST: 0, MST: 0
 BPDU Received      :158
        TCN: 0, Config: 158, RST: 0, MST: 0
```

Step 2 **Control root bridge election.**

Run the **display stp** command to view information about the root bridge.

```
<S1>display  stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge        :0    .4c1f-cc45-aace
Bridge Times       :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC     :0    .4c1f-cc45-aace / 0
CIST RegRoot/IRPC   :0    .4c1f-cc45-aace / 0
CIST RootPortId    :0.0
BPDU-Protection    :Disabled
CIST Root Type     :Primary root
TC or TCN received  :108
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:9m:23s
```

```
……output omit……


<S2>display stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge        :4096 .4c1f-cc45-aacc
Bridge Times       :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC     :0    .4c1f-cc45-aace / 20000
CIST RegRoot/IRPC   :4096 .4c1f-cc45-aacc / 0
CIST RootPortId    :128.9
BPDU-Protection    :Disabled
CIST Root Type     :Secondary root
TC or TCN received  :55
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:9m:30s
……output omit……
```

Configure S1 as the root bridge and S2 as the backup root bridge using priority values. The device with the same value for the **CIST Bridge** and **CIST Root/ERPC** is the root bridge. A smaller bridge priority value indicates a higher bridge priority. Change the priorities of S1 and S2 to 8192 and 4096 respectively so that S2 becomes the root bridge.

```
[S1]undo stp root
[S1]stp priority 8192

[S2]undo stp root
[S2]stp priority 4096
```

Run the **display stp** command to view information about the new root bridge.

```
<S1>display stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge        :8192 .4c1f-cc45-aace
Bridge Times       :Hello 2s MaxAge 20s FwDly 15s 0
CIST Root/ERPC     :4096 .4c1f-cc45-aacc / 20000
CIST RegRoot/IRPC   :8192 .4c1f-cc45-aace / 0
CIST RootPortId    :128.9
BPDU-Protection    :Disabled
TC or TCN received  :143
TC count per hello  :0
```

```
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:0m:27s
……output omit……
<S2>display  stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge         :4096 .4c1f-cc45-aacc
Bridge Times        :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC      :4096 .4c1f-cc45-aacc / 0
CIST RegRoot/IRPC   :4096 .4c1f-cc45-aacc / 0
CIST RootPortId     :0.0
BPDU-Protection     :Disabled
TC or TCN received  :55
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:14m:7s
……output omit……
```

The highlighted lines in the preceding information indicate that S2 has become the new root bridge.

Shut down interfaces Gigabit Ethernet 0/0/9 and Gigabit Ethernet 0/0/10 on S2 to isolate S2.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]shutdown
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]shutdown

[S1]display  stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge         :8192 .4c1f-cc45-aace
Bridge Times        :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC      :8192 .4c1f-cc45-aace / 0
CIST RegRoot/IRPC   :8192 .4c1f-cc45-aace / 0
CIST RootPortId     :0.0
BPDU-Protection     :Disabled
TC or TCN received  :146
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:0m:11s
```

```
……output omit……
```

The highlighted lines in the preceding information indicate that S1 becomes the root bridge when S2 is faulty.

Re-enable the interfaces that have been disabled on S2.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]undo shutdown
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]undo shutdown

<S1>display stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge         :8192 .4c1f-cc45-aace
Bridge Times        :Hello 2s MaxAge 20s FwDly 15s 0
CIST Root/ERPC      :4096 .4c1f-cc45-aacc / 20000
CIST RegRoot/IRPC   :8192 .4c1f-cc45-aace / 0
CIST RootPortId     :128.9
BPDU-Protection     :Disabled
TC or TCN received  :143
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:0m:27s
……output omitted……

<S2>display  stp
-------[CIST Global Info][Mode STP]-------
CIST Bridge         :4096 .4c1f-cc45-aacc
Bridge Times        :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC      :4096 .4c1f-cc45-aacc / 0
CIST RegRoot/IRPC   :4096 .4c1f-cc45-aacc / 0
CIST RootPortId     :0.0
BPDU-Protection     :Disabled
TC or TCN received  :55
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:14m:7s
……output omitted……
```

The highlighted lines in the preceding information indicate that S2 has been restored and has become the root bridge once again.

Step 3 **Control root port election.**

Run the **display stp brief** command on S1 to view the roles of the interfaces.

```
<S1>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    ROOT    FORWARDING  NONE
    0       GigabitEthernet0/0/10   ALTE    DISCARDING  NONE
```

The preceding information shows that G0/0/9 is the root port and G0/0/10 is the alternate port. You can change port priorities so that port interface G0/0/10 will become the root port and G0/0/9 will become the alternate port.

Change priorities of G0/0/9 and G0/0/10 on S2.

The default port priority is 128. A larger port priority value indicates a lower priority. The priorities of G0/0/9 and G0/0/10 on S2 are set to 32 and 16; therefore, G0/0/10 on S1 becomes the root port.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]stp port priority 32
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]stp port priority 16
```

Note that the port priorities are changed on S2, not S1.

```
<S2>display stp interface  GigabitEthernet 0/0/9
----[CIST][Port9(GigabitEthernet0/0/9)][FORWARDING]----
 Port Protocol      :Enabled
 Port Role          :Designated Port
 Port Priority      :32
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port   :4096.4c1f-cc45-aacc / 32.9
 Port Edged         :Config=default / Active=disabled
 Point-to-point     :Config=auto / Active=true
 Transit Limit      :147 packets/hello-time
 Protection Type    :None
 Port STP Mode      :STP
 Port Protocol Type  :Config=auto / Active=dot1s
```

```
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 20
 TC or TCN send      :22
 TC or TCN received  :1
 BPDU Sent          :164
         TCN: 0, Config: 164, RST: 0, MST: 0
 BPDU Received       :2
         TCN: 1, Config: 1, RST: 0, MST: 0

<S2>display stp interface GigabitEthernet 0/0/10
----[CIST][Port10(GigabitEthernet0/0/10)][FORWARDING]----
 Port Protocol       :Enabled
 Port Role          :Designated Port
 Port Priority       :16
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port   :4096.4c1f-cc45-aacc / 16.10
 Port Edged         :Config=default / Active=disabled
 Point-to-point      :Config=auto / Active=true
 Transit Limit       :147 packets/hello-time
 Protection Type     :None
 Port STP Mode       :STP
 Port Protocol Type  :Config=auto / Active=dot1s
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 20
 TC or TCN send      :35
 TC or TCN received  :1
 BPDU Sent          :183
         TCN: 0, Config: 183, RST: 0, MST: 0
 BPDU Received       :2
         TCN: 1, Config: 1, RST: 0, MST: 0
```

Run the **display stp brief** command on S1 to view the role of the interfaces.

```
<S1>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    ALTE    DISCARDING  NONE
    0       GigabitEthernet0/0/10   ROOT    FORWARDING  NONE
```

The highlighted lines in the preceding information indicate that G0/0/10 on S1 has become the root port and G0/0/9 has become the alternate port.

Shut down G0/0/10 on S1 and view the port roles.

```
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]shutdown
<S1>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    ROOT    FORWARDING  NONE
```

The highlighted line in the preceding information indicates that G0/0/9 has become the root port. Resume the default priorities of G0/0/9 and G0/0/10 on S2 and re-enable the shutdown interfaces on S1.

```
[S2]interface GigabitEthernet 0/0/9
[S2-GigabitEthernet0/0/9]undo stp port priority
[S2-GigabitEthernet0/0/9]quit
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]undo stp port priority


[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]undo shutdown
```

Run the **display stp brief** and **display stp interface** command on S1 to view the roles of interfaces.

```
<S1>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    ROOT    FORWARDING  NONE
    0       GigabitEthernet0/0/10   ALTE    DISCARDING  NONE


[S1]display stp interface GigabitEthernet 0/0/9
----[CIST][Port9(GigabitEthernet0/0/9)][FORWARDING]----
 Port Protocol      :Enabled
 Port Role          :Root Port
 Port Priority      :128
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port   :4096.4c1f-cc45-aacc / 128.9
 Port Edged         :Config=default / Active=disabled
 Point-to-point     :Config=auto / Active=true
 Transit Limit      :147 packets/hello-time
 Protection Type    :None
 Port STP Mode      :STP
 Port Protocol Type  :Config=auto / Active=dot1s
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 0
 TC or TCN send     :4
```

```
 TC or TCN received  :90
 BPDU Sent          :5
        TCN: 4, Config: 1, RST: 0, MST: 0
 BPDU Received      :622
        TCN: 0, Config: 622, RST: 0, MST: 0

[S1]display stp interface GigabitEthernet 0/0/10
----[CIST][Port10(GigabitEthernet0/0/10)][DISCARDING]----
 Port Protocol      :Enabled
 Port Role          :Alternate Port
 Port Priority      :128
 Port Cost(Dot1T )   :Config=auto / Active=20000
 Designated Bridge/Port   :4096.4c1f-cc45-aacc / 128.10
 Port Edged         :Config=default / Active=disabled
 Point-to-point     :Config=auto / Active=true
 Transit Limit      :147 packets/hello-time
 Protection Type    :None
 Port STP Mode      :STP
 Port Protocol Type  :Config=auto / Active=dot1s
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 0
 TC or TCN send     :3
 TC or TCN received  :90
 BPDU Sent          :4
        TCN: 3, Config: 1, RST: 0, MST: 0
 BPDU Received      :637
        TCN: 0, Config: 637, RST: 0, MST: 0
```

The greyed line in the preceding information indicates that G0/0/9 and G0/0/10 cost is 20000 by default.

Change the cost of G0/0/9 to 200000 on S1.

```
[S1]interface GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]stp cost 200000
```

Run the **display stp brief** and **display stp interface** command on S1 to view the roles of interfaces.

```
<S1>display stp interface GigabitEthernet 0/0/9
----[CIST][Port9(GigabitEthernet0/0/9)][DISCARDING]----
 Port Protocol      :Enabled
 Port Role          :Alternate Port
 Port Priority      :128
```

```
 Port Cost(Dot1T )   :Config=200000 / Active=200000
 Designated Bridge/Port   :4096.4c1f-cc45-aacc / 128.9
 Port Edged         :Config=default / Active=disabled
 Point-to-point      :Config=auto / Active=true
 Transit Limit       :147 packets/hello-time
 Protection Type     :None
 Port STP Mode       :STP
 Port Protocol Type  :Config=auto / Active=dot1s
 BPDU Encapsulation  :Config=stp / Active=stp
 PortTimes          :Hello 2s MaxAge 20s FwDly 15s RemHop 0
 TC or TCN send      :4
 TC or TCN received  :108
 BPDU Sent          :5
         TCN: 4, Config: 1, RST: 0, MST: 0
 BPDU Received       :818
         TCN: 0, Config: 818, RST: 0, MST: 0

<S1>display stp brief
    MSTID   Port                    Role    STP State   Protection
    0       GigabitEthernet0/0/9    ALTE    DISCARDING  NONE
    0       GigabitEthernet0/0/10   ROOT    FORWARDING  NONE
```

The highlighted lines in the preceding information indicates that G0/0/10 has become the root port.

Final Configuration

```
<S1>display current-configuration
#
!Software Version V100R006C00SPC800
 sysname S1
#
 stp mode stp
 stp instance 0 priority 8192
#
interface GigabitEthernet0/0/9
 stp instance 0 cost 200000
#
interface GigabitEthernet0/0/10
#
user-interface con 0
user-interface vty 0 4
```

```
#
return

<S2>display current-configuration
#
!Software Version V100R006C00SPC800
 sysname S2
#
 stp mode stp
 stp instance 0 priority 4096
#
interface GigabitEthernet0/0/9
#
interface GigabitEthernet0/0/10
#
user-interface con 0
user-interface vty 0 4
#
return

<S3>display current-configuration
#
!Software Version V100R006C00SPC800
 sysname S3
#
interface Ethernet0/0/1
 shutdown
#
interface Ethernet0/0/13
 shutdown
#
interface Ethernet0/0/23
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return


<S4>display current-configuration
#
!Software Version V100R006C00SPC800
```

```
 sysname S4
#
interface Ethernet0/0/14
 shutdown
#
interface Ethernet0/0/24
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return
```

Lab 3-2 Configuring RSTP

Learning Objectives

As a result of this lab section, you should achieve the following tasks:

- Enable and disable RSTP.
- Configuration of an edge port.
- Configuration of RSTP BPDU protection.
- Configuration of RSTP loop protection.

Topology

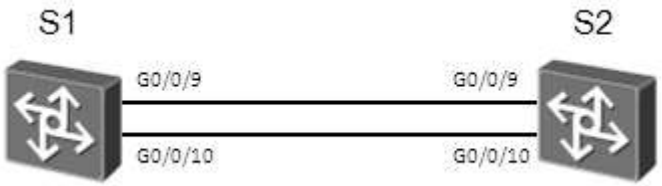


Figure 3.2 RSTP topology

Scenario

Assume that you are a network administrator of a company. The company network consists of two layers: core layer and access layer. The network uses a redundancy design. RSTP will be used to prevent loops. You can configure features to speed up RSTP route convergence at the edge network and configure RSTP protection function.

Tasks

Step 1 Preparing the environment.

If you are starting this section with a non-configured device, begin here

and then move to step 3. For those continuing from previous labs, begin at step 2.

Irrelevant interfaces must be disabled to ensure test result accuracy.

Shut down port interfaces Ethernet 0/0/1 on S3, Ethernet 0/0/13 and Ethernet 0/0/23 on S3, as well as Ethernet 0/0/14 and Ethernet 0/0/24 on S4 before starting STP configuration. Ensure that the devices start without any configuration files. If STP is disabled, run the stp enable command to enable STP.

```
<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S1

<Quidway>system-view
Enter system view, return user view with Ctrl+Z.
[Quidway]sysname S2

<Quidway>system-view
[Quidway]sysname R3
[R3]interface Ethernet 0/0/1
[R3-Ethernet0/0/1]shutdown
[R3-Ethernet0/0/1]quit
[R3]interface Ethernet 0/0/13
[R3-Ethernet0/0/13]shutdown
[R3-Ethernet0/0/13]quit
[R3]interface Ethernet 0/0/23
[R3-Ethernet0/0/23]shutdown

<Quidway>system-view
[Quidway]sysname S4
[S4]inter Ethernet 0/0/14
[S4-Ethernet0/0/14]shutdown
[S4-Ethernet0/0/14]quit
[S4]interface Ethernet 0/0/24
[S4-Ethernet0/0/24]shutdown
```

Step 2 **Clean up the previous configuration.**

Remove the configured STP priority from S1 and S2, and assigned cost on S1.

```
[S1]undo stp priority
```

```
[S1]inter GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]undo stp cost

[S2]undo stp priority
```

Step 3 **Configure RSTP and verify the RSTP configuration.**

Configure S1 and S2 to use RSTP as the spanning tree protocol.

```
[S1]stp mode rstp

[S2]stp mode rstp
```

Run the **display stp** command to view brief information about RSTP.

```
[S1]display  stp
-------[CIST Global Info][Mode RSTP]-------
CIST Bridge        :32768.4c1f-cc45-aace
Bridge Times       :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC     :32768.4c1f-cc45-aacc / 20000
CIST RegRoot/IRPC   :32768.4c1f-cc45-aace / 0
CIST RootPortId    :128.9
BPDU-Protection    :Disabled
TC or TCN received  :28
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:11m:1s
……output omitted……

[S2]display  stp
-------[CIST Global Info][Mode RSTP]-------
CIST Bridge        :32768.4c1f-cc45-aacc
Bridge Times       :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC     :32768.4c1f-cc45-aacc / 0
CIST RegRoot/IRPC   :32768.4c1f-cc45-aacc / 0
CIST RootPortId    :0.0
BPDU-Protection    :Disabled
TC or TCN received  :14
TC count per hello  :0
STP Converge Mode   :Normal
Share region-configuration :Enabled
Time since last TC  :0 days 0h:12m:23s
……output omitted……
```

Step 4 **Configure an edge port.**

Configure ports connected to the user terminals as edge ports. An edge port can transition to the forwarding state without participating in the RSTP calculation. In this example, interface Gigabit Ethernet 0/0/4 on S1 and S2 connect to a router and can be configured as edge ports.

```
[S1]interface GigabitEthernet 0/0/4
[S1-GigabitEthernet0/0/4]stp edged-port enable

[S2]interface GigabitEthernet 0/0/4
[S2-GigabitEthernet0/0/4]stp edged-port enable
```

Step 5 **Configure BPDU protection.**

Edge ports are directly connected to user terminal and will not receive BPDUs. Attackers may send pseudo BPDUs to attack the switching device. If the edge ports receive the BPDUs, the switching device configures the edge ports as non-edge ports and triggers a new spanning tree calculation. Network flapping then occurs. BPDU protection can be used to protect switching devices against malicious attacks.

Configure BPDU protection on both S1 and S2.

```
[S1]stp bpdu-protection

[S2]stp bpdu-protection
```

Run the **display stp brief** command to view the port protection.

```
<S1>display  stp brief
 MSTID  Port                      Role  STP State     Protection
   0    GigabitEthernet0/0/4       DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/9       ROOT  FORWARDING      NONE
   0    GigabitEthernet0/0/10      ALTE  DISCARDING      NONE

<S2>display  stp brief
 MSTID  Port                      Role  STP State     Protection
   0    GigabitEthernet0/0/4       DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/9       DESI  FORWARDING      NONE
   0    GigabitEthernet0/0/10      DESI  FORWARDING      NONE
```

After the configuration is complete, interface Gigabit Ethernet 0/0/4 on S1 and S2 shows as supporting BPDU protection.

Step 6 **Configure Loop protection.**

On a network running RSTP, a switching device maintains the root port status and status of alternate ports by receiving BPDUs from an upstream switching device. If the switching device cannot receive BPDUs from the upstream device because of link congestion or unidirectional-link failure, the switching device re-selects a root port. The original root port becomes a designated port and the original discarding ports change to the Forwarding state. This switching may cause network loops, which can be mitigated by configuring loop protection.

Configure loop protection on both the root port and the alternate port.

```
[S1]display  stp brief
 MSTID  Port                    Role  STP State     Protection
   0    GigabitEthernet0/0/4      DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/9      ROOT  FORWARDING      NONE
   0    GigabitEthernet0/0/10     ALTE  DISCARDING      NONE
```

G0/0/9 and G0/0/10 on S1 are now the root port and alternate port. Configure loop protection on these two ports.

```
[S1]interface GigabitEthernet 0/0/9
[S1-GigabitEthernet0/0/9]stp loop-protection
[S1-GigabitEthernet0/0/9]quit
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]stp loop-protection
```

Run the **display stp brief** command to view the port protection.

```
<S1>display  stp brief
 MSTID  Port                    Role  STP State     Protection
   0    GigabitEthernet0/0/4      DESI  FORWARDING      BPDU
   0    GigabitEthernet0/0/9      ROOT  FORWARDING      LOOP
   0    GigabitEthernet0/0/10     ALTE  DISCARDING      LOOP
```

Since S2 is root, all the ports are designated ports and therefore do not need to configure loop protection. After completing the configuration, you may wish to set S1 as the root, and configure loop protection on the root port and alternate port of S2 using the same process as with S1.

Final Configuration

```
<S1>display current-configuration
#
!Software Version V100R006C00SPC800
 sysname S1
#
 stp mode rstp
 stp bpdu-protection
#
interface GigabitEthernet0/0/4
 stp edged-port enable
#
interface GigabitEthernet0/0/9
 stp loop-protection
#
interface GigabitEthernet0/0/10
 stp loop-protection
#
user-interface con 0
user-interface vty 0 4
#
return

<S2>display current-configuration
#
!Software Version V100R006C00SPC800
 sysname S2
#
 stp mode rstp
 stp bpdu-protection
#
interface GigabitEthernet0/0/4
 stp edged-port enable
#
user-interface con 0
user-interface vty 0 4
#
Return

<S3>display current-configuration
#
```

```
!Software Version V100R006C00SPC800
 sysname S3
#
interface Ethernet0/0/1
 shutdown
#
interface Ethernet0/0/13
 shutdown
#
interface Ethernet0/0/23
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return

<S4>dis current-configuration
#
!Software Version V100R006C00SPC800
 sysname S4
#
interface Ethernet0/0/14
 shutdown
#
interface Ethernet0/0/24
 shutdown
#
user-interface con 0
user-interface vty 0 4
#
return
```